6. QR factorization

- QR factorization
- QR via Gram-Schmidt
- modified Gram-Schmidt
- pivoted QR factorization
- Householder algorithm

QR factorization

any $A \in \mathbb{R}^{m \times n}$ with linearly independent columns admits the factorization:

$$A = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1n} \\ 0 & R_{22} & \cdots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R_{nn} \end{bmatrix} = QR$$

- Q is $m \times n$ with orthonormal columns $(Q^T Q = I)$
- R is $n \times n$, upper triangular, with nonzero diagonal elements (invertible)
- most definitions assume $R_{ii} > 0$ making Q and R unique:

  if $R_{ii} < 0$, we can switch the signs of $R_{ii}, \ldots, R_{in}$ and vector q_i
- called *QR factorization* (also known as *reduced* or *thin* QR factorization)
- several algorithms (in MATLAB: `[Q,R]=qr(A,0)`), including Gram-Schmidt

Pseudo-inverse via QR factorization

pseudo-inverse of A with linearly independent columns with $A = QR$ is

$$\begin{aligned}
A^\dagger &= (A^T A)^{-1} A^T \\
&= \big((QR)^T (QR)\big)^{-1} (QR)^T \\
&= (R^T Q^T Q R)^{-1} R^T Q^T \\
&= (R^T R)^{-1} R^T Q^T \quad (Q^T Q = I) \\
&= R^{-1} R^{-T} R^T Q^T \quad (R \text{ is nonsingular}) \\
&= R^{-1} Q^T
\end{aligned}$$

- for square nonsingular A this is the inverse: $A^{-1} = (QR)^{-1} = R^{-1} Q^T$
- pseudo-inverse of A with linearly independent rows with $A^T = \hat{Q}\hat{R}$ is

$$A^\dagger = A^T (A A^T)^{-1} = \hat{Q}\hat{R}^{-T}$$

Range and QR factorization

suppose A has linearly independent columns with QR factorization $A = QR$

- Q has the same range as A:

$$\begin{aligned} y \in \mathrm{range}(A) &\iff y = Ax \text{ for some } x \\ &\iff y = QRx \text{ for some } x \\ &\iff y = Qz \text{ for some } z \\ &\iff y \in \mathrm{range}(Q) \end{aligned}$$

- columns of Q are orthonormal *basis* for $\mathrm{range}(A)$:

  they are linearly independent and span$(q_1, \ldots, q_n) = \mathrm{range}(A)$

Full QR factorization

the QR factorization of $A \in \mathbb{R}^{m \times n}$ with lin. indep. columns is usually defined as

$$A = \begin{bmatrix} Q & Q_0 \end{bmatrix} \begin{bmatrix} R \\ 0 \end{bmatrix}$$

- $A = QR$ with same Q, R as before
- $[Q \quad Q_0]$ is $m \times m$ and orthogonal
- Q_0 has size $m \times (m - n)$ and zero block has size $(m - n) \times n$
- several algorithms, including Householder; in MATLAB: `[Q,R]=qr(A)`
- called *full QR factorization* or *QR decomposition*

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Matrix form of Gram-Schmidt

let A be an $m \times n$ matrix with linearly independent columns

- running Gram-Schmidt on A produces orthonormal vectors $q_1, \ldots, q_n$
- we know from Gram-Schmidt algorithm that

$$\begin{aligned} a_k &= (q_1^T a_k) q_1 + \cdots + (q_{k-1}^T a_k) q_{k-1} + \|\tilde{q}_k\| q_k \\ &= R_{1k} q_1 + \cdots + R_{k-1,k} q_{k-1} + R_{kk} q_k \end{aligned}$$

  where $R_{ij} = q_i^T a_j$ and $R_{ii} = \|\tilde{q}_i\| > 0$

- expressing this for each $k = 1, \ldots, n$:

$$\begin{aligned} a_1 &= R_{11} q_1 \\ a_2 &= R_{12} q_1 + R_{22} q_2 \\ &\vdots \\ a_n &= R_{1n} q_1 + \cdots + R_{nn} q_n \end{aligned} \qquad A = \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & \ldots & R_{1n} \\ 0 & R_{22} & \ldots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & R_{nn} \end{bmatrix}$$

QR factorization via Gram-Schmidt

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given: $m \times n$ matrix A with linearly independent columns $a_1, \ldots, a_n$
set $q_1 = a_1/\|a_1\|$ and $R_{11} = \|a_1\|$

for $k = 2, \ldots, n$

1. $\tilde{q}_k = a_k$
2. **for** $j = 1, \ldots, k-1$
   $R_{jk} = q_j^T a_k$
   $\tilde{q}_k = \tilde{q}_k - R_{jk} q_j$
3. set
   $R_{kk} = \|\tilde{q}_k\|$
   $q_k = \tilde{q}_k / R_{kk}$

- Q and R is generated column by column
- **complexity:** $2mn^2$ flops

Gram-Schmidt algorithm after k steps

- after k steps we have a partial QR factorization

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_k \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & \cdots & q_k \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1k} \\ 0 & R_{22} & \cdots & R_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R_{kk} \end{bmatrix}$$

- this is the QR factorization for the first k columns of A
- columns $q_1, \ldots, q_k$ are orthonormal
- diagonal elements $R_{11}, R_{22}, \ldots, R_{kk}$ are positive
- columns $q_1, \ldots, q_k$ have the same span as $a_1, \ldots, a_k$
- in step k of the algorithm we compute

$$q_k, \; R_{1k}, \ldots, R_{kk}$$

Example

$$A = [a_1 \;\; a_2 \;\; a_3] = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix}$$

- $k = 1$ (first column of Q and R)

$$q_1 = a_1/\|a_1\| = (-1/2, 1/2, -1/2, 1/2), \quad R_{11} = \|a_1\| = 2$$

- $k = 2$ (second column of Q and R)

$$R_{12} = q_1^T a_2 = 4, \quad \tilde{q}_2 = a_2 - R_{12} q_1 = (1, 1, 1, 1)$$

normalizing, we get

$$R_{22} = \|\tilde{q}_2\| = 2, \quad q_2 = \tilde{q}_2/R_{22} = (1/2, 1/2, 1/2, 1/2)$$

Example

- $k = 3$ (third column of Q and R)

$$R_{13} = q_1^T a_3 = 2, \quad R_{23} = q_2^T a_3 = 8$$
$$\tilde{q}_3 = a_3 - R_{13} q_1 - R_{23} q_2 = (-2, -2, 2, 2)$$

normalizing, we get

$$R_{33} = \|\tilde{q}_3\| = 4, \quad q_3 = \tilde{q}_3 / R_{33} = (-1/2, -1/2, 1/2, 1/2)$$

Final factorization

$$\begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ 0 & 0 & R_{33} \end{bmatrix}$$
$$= \begin{bmatrix} -1/2 & 1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$

Numerical instability of G-S

consider the following MATLAB implementation of the G-S algorithm

```
[m, n] = size(A);
Q = zeros(m,n);
R = zeros(n,n);
for k = 1:n
R(1:k-1,k) = Q(:,1:k-1)' * A(:,k);
qtilde = A(:,k) - Q(:,1:k-1) * R(1:k-1,k);
R(k,k) = norm(qtilde);
Q(:,k) = qtilde / R(k,k);
end;
```

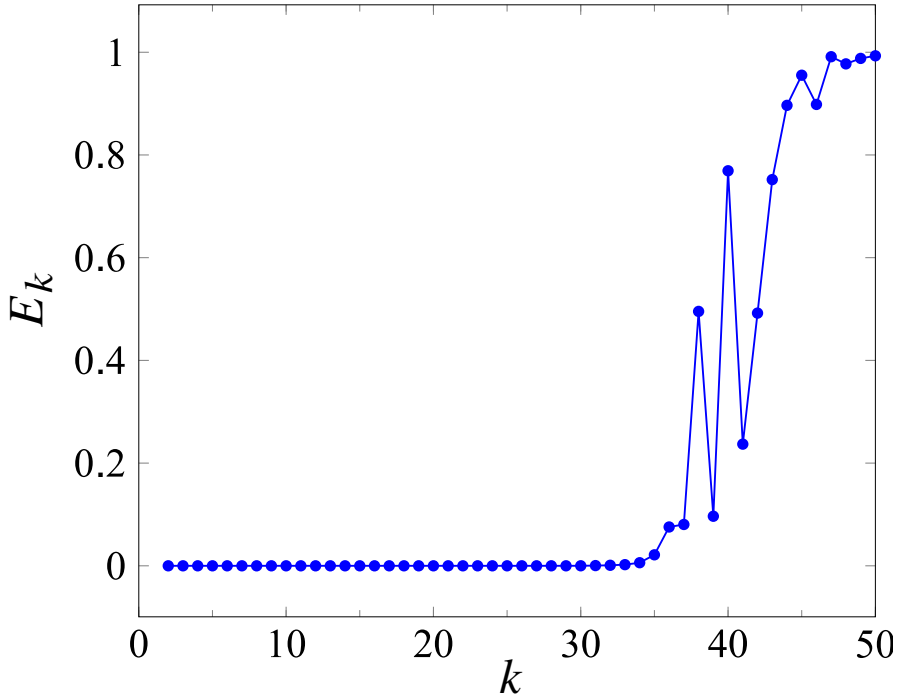
- we apply this to a square matrix A of size $m = n = 50$
- A is constructed as $A = USV$ with U, V orthogonal, S diagonal with

$$S_{ii} = 10^{-10(i-1)/(n-1)}, \quad i = 1, \ldots, n$$

Numerical instability of G-S

plot shows deviation from orthogonality between q_k and previous columns

$$E_k = \max_{1 \le i < k} |q_i^T q_k|, \quad k = 2, \ldots, n$$



loss of orthogonality is due to rounding error

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Modified Gram-Schmidt algorithm

a variation of the classical Gram-Schmidt algorithm for QR factorization

$$\begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1n} \\ 0 & R_{22} & \cdots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R_{nn} \end{bmatrix}$$

of a matrix with linearly independent columns

- mathematically equivalent to classical G-S
- has better numerical properties than classical Gram-Schmidt algorithm
- computes Q column by column, R row by row

Gram-Schmidt orthogonalization

recall G-S: for $k = 1, \ldots, n$

$$\tilde{q}_k = a_k - \sum_{j=1}^{k-1} (q_j^T a_k)\, q_j \quad (R_{jk} = q_j^T a_k)$$

$$R_{kk} = \|\tilde{q}_k\|, \qquad q_k = \frac{\tilde{q}_k}{R_{kk}}$$

Projection interpretation

- recall the Gram-Schmidt computes the projection of a_k onto

  $$\text{span}\{a_1, \ldots, a_{k-1}\} = \text{span}\{q_1, \ldots, q_{k-1}\}$$

  and subtracts it (for $k \geq 2$):

  $$\tilde{q}_k = (I - \sum_{j=1}^{k-1} q_j q_j^T) a_k = (I - [q_1 \cdots q_{k-1}][q_1 \cdots q_{k-1}]^T) a_k$$

- this is the orthogonal projector onto $\text{span}\{a_1, \ldots, a_{k-1}\}^\perp$

Modified Gram-Schmidt: idea

- the orthogonal projector can be written as a product of elementary projections

$$I - \sum_{j=1}^{k-1} q_j q_j^T = (I - q_{k-1} q_{k-1}^T) \cdots (I - q_1 q_1^T)$$

- thus the residual vector for column k can be written as

$$\tilde{q}_k = (I - q_{k-1} q_{k-1}^T) \cdots (I - q_1 q_1^T) a_k, \qquad k \geq 2$$

- after computing $q_k = \frac{\tilde{q}_k}{\|\tilde{q}_k\|}$, we orthogonalize all remaining columns $j > k$:

$$R_{kj} = q_k^T \tilde{q}_j^{(k-1)} \quad \text{(gives } R_{kj} = q_k^T a_j \text{ due to orthogonality)}$$
$$\tilde{q}_j^{(k)} = (I - q_k q_k^T) \tilde{q}_j^{(k-1)} = \tilde{q}_j^{(k-1)} - R_{kj} q_k, \quad j = k+1, \ldots, n$$

with $\tilde{q}_{k+1} = \tilde{q}_{k+1}^{(k)}$; we start with $\tilde{q}_j^{(0)} = a_j$

- this can be formulated into the modified G-S algorithm next

Modified Gram-Schmidt algorithm

---

given: $m \times n$ matrix A with linearly independent columns $a_1, \ldots, a_n$
set $\tilde{Q} = A$

for $k = 1, \ldots, n$

1. set

   $R_{kk} = \|\tilde{q}_k\|$
   $q_k = \tilde{q}_k / R_{kk}$
2. **for** $j = k + 1, \ldots, n$

   $R_{kj} = q_k^T \tilde{q}_j$
   $\tilde{q}_j = \tilde{q}_j - R_{kj} q_k$

- R is generated *row* by *row* (at step k, we compute the kth row $R_{k,k:n}$)
- **complexity:** $2mn^2$ flops

Modified Gram-Schmidt in matrix-vector form

given: $m \times n$ matrix A with linearly independent columns $a_1, \ldots, a_n$
set $\tilde{Q}_0 = A$

for $k = 1$ to n,

1. compute $R_{kk} = \|\tilde{q}_k\|$ and $q_k = (1/R_{kk})\tilde{q}_k$ where $\tilde{q}_k$ is the first column of $\tilde{Q}_{k-1}$
2. compute

$$[R_{k,k+1} \cdots R_{kn}] = q_k^T B, \quad \tilde{Q}_k = B - q_k [R_{k,k+1} \cdots R_{kn}]$$

where B is $\tilde{Q}_{k-1}$ with the first column removed

MATLAB implementation (`Q(:,k:n)` used to store $\tilde{Q}_{k-1}$)

```
Q = A; R = zeros(n,n);
for k = 1:n
R(k,k) = norm(Q(:,k));
Q(:,k) = Q(:,k) / R(k,k);
R(k,k+1:n) = Q(:,k)' * Q(:,k+1:n);
Q(:,k+1:n) = Q(:,k+1:n) - Q(:,k) * R(k,k+1:n);
end;
```

Modified Gram-Schmidt algorithm after k steps

after k steps ($k = 1, \ldots, n$), the algorithm has computed a partial QR factorization

$$
\begin{aligned}
A &= [a_1 \cdots a_k \mid a_{k+1} \cdots a_n] \\
&= \left[q_1 \cdots q_k \mid \tilde{Q}_k\right] \left[\begin{array}{ccc|ccc}
R_{11} & \cdots & R_{1k} & R_{1,k+1} & \cdots & R_{1n} \\
\vdots & \ddots & \vdots & \vdots & & \vdots \\
0 & \cdots & R_{kk} & R_{k,k+1} & \cdots & R_{kn} \\
\hline
 & 0 & & & I & \\
\end{array}\right]
\end{aligned}
$$

- $q_1, \ldots, q_k$ are orthonormal vectors; $R_{11}, \ldots, R_{kk}$ are positive
- columns of $\tilde{Q}_k$ are projection of $a_{k+1}, \ldots, a_n$ on span$(q_1, \ldots, q_k)^\perp$
- the factorization starts with $\tilde{Q}_0 = A$ and is complete when $k = n$
- in step k, we compute

$$q_k, \;\; R_{kk}, \;\; R_{k,k+1}, \ldots, R_{kn}, \;\; \tilde{Q}_k$$

Example

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix}$$

Step 1: first column of Q, first row of R

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \left[\begin{array}{r|cc} -1/2 & 1 & 2 \\ 1/2 & 1 & 2 \\ -1/2 & 1 & 6 \\ 1/2 & 1 & 6 \end{array}\right] \left[\begin{array}{c|cc} 2 & 4 & 2 \\ \hline 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{c|c} q_1 & \tilde{Q}_2 \end{array}\right] \left[\begin{array}{c|c} R_{11} & R_{1,2:3} \\ \hline 0 & I \end{array}\right]$$

Example

Step 2: second column of Q, second row of R

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \left[\begin{array}{cc|c} -1/2 & 1/2 & -2 \\ 1/2 & 1/2 & -2 \\ -1/2 & 1/2 & 2 \\ 1/2 & 1/2 & 2 \end{array}\right] \left[\begin{array}{cc|c} 2 & 4 & 2 \\ 0 & 2 & 8 \\ \hline 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{cc|c} q_1 & q_2 & \tilde{Q}_3 \end{array}\right] \left[\begin{array}{cc|c} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ \hline 0 & 0 & 1 \end{array}\right]$$

Step 3: third column of Q, third row of R

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \begin{bmatrix} -1/2 & 1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ 0 & 0 & R_{33} \end{bmatrix}$$

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QR factorization with column pivoting

A is an $m \times n$ matrix (may be wide or have linearly dependent columns)

QR factorization with column pivoting (column reordering)

$$A = QRP^T$$

- Q is $m \times r$ with orthonormal columns
- R is $r \times n$, leading $r \times r$ submatrix is upper triangular with positive diagonal:

$$R = \left[\begin{array}{c|c} R_1 & R_2 \end{array}\right] = \left[\begin{array}{cccc|ccc} R_{11} & R_{12} & \cdots & R_{1r} & R_{1,r+1} & \cdots & R_{1n} \\ 0 & R_{22} & \cdots & R_{2r} & R_{2,r+1} & \cdots & R_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & R_{rr} & R_{r,r+1} & \cdots & R_{rn} \end{array}\right]$$

- can be chosen so that $R_{11} \geq R_{22} \geq \cdots \geq R_{rr} > 0$
- P is an $n \times n$ permutation matrix
- r is the rank of A: gives full rank factorization $A = BC$ with $B = Q$, $C = RP^T$

Interpretation

- columns of $AP = QR$ are the columns of A in a different order
- the columns are divided in two groups:

$$AP = [\hat{A}_1 \;\; \hat{A}_2] = Q[R_1 \;\; R_2], \quad \hat{A}_1 \text{ is } m \times r,\; R_1 \text{ is } r \times r$$

- $\hat{A}_1 = QR_1$ is $m \times r$ with linearly independent columns:

$$\hat{A}_1 x = QR_1 x = 0 \quad \Longrightarrow \quad R_1^{-1} Q^T \hat{A}_1 x = x = 0$$

- $\hat{A}_2 = QR_2$ is $m \times (n - r)$: columns are linear combinations of columns of $\hat{A}_1$

$$\hat{A}_2 = QR_2 = \hat{A}_1 R_1^{-1} R_2$$

the factorization provides two useful bases for $\mathrm{range}(A)$

- columns of Q are an orthonormal basis
- columns of $\hat{A}_1$ are a basis selected from the columns of A

Dimension of nullspace

if A is $m \times n$ then

$$\dim(\mathrm{null}(A)) = n - \mathrm{rank}(A)$$

- $\dim(\mathrm{null}(A))$ is known as the *nullity* of the matrix
- we show this by constructing a basis containing $n - \mathrm{rank}(A)$ vectors

Basis for nullspace: a basis for the nullspace of A is given by the columns of

$$P \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}$$

where P, R_1, R_2 are the matrices in the pivoted QR factorization

$$AP = Q \begin{bmatrix} R_1 & R_2 \end{bmatrix}$$

- P is a $n \times n$ permutation matrix
- Q is $m \times r$ with orthonormal columns, where $r = \mathrm{rank}(A)$
- R_1 is $r \times r$ upper triangular and nonsingular, R_2 is $r \times (n - r)$

Proof

- x is in the nullspace of A if and only if $y = P^T x$ is in the nullspace of AP
- $y = (y_1, y_2)$ is in the nullspace of AP if and only if

$$
\begin{aligned}
APy = 0 &\iff Q \begin{bmatrix} R_1 & R_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0 \\
&\iff \begin{bmatrix} R_1 & R_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0 \qquad (Q \text{ has orthonormal columns}) \\
&\iff \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} y_2 \qquad (R_1 \text{ nonsingular})
\end{aligned}
$$

- therefore, x is in the nullspace of A if and only if it is in the range of

$$P \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}$$

- the columns of this matrix are linearly independent, so they are a basis for

$$\mathrm{range}\left(P \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} \right) = \mathrm{null}(A)$$

Modified Gram-Schmidt with column pivoting

with minor changes the modified GS algorithm computes the pivoted factorization

$$AP = \begin{bmatrix} q_1 & q_2 & \cdots & q_r \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1r} & R_{1,r+1} & \cdots & R_{1n} \\ 0 & R_{22} & \cdots & R_{1r} & R_{1,r+1} & \cdots & R_{1n} \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & R_{rr} & R_{r,r+1} & \cdots & R_{rn} \end{bmatrix}$$

- partial factorization after k steps

$$AP_k = \begin{bmatrix} q_1 \cdots q_k \mid \tilde{Q}_k \end{bmatrix} \left[\begin{array}{ccc|ccc} R_{11} & \cdots & R_{1k} & R_{1,k+1} & \cdots & R_{1n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & R_{kk} & R_{k,k+1} & \cdots & R_{kn} \\ \hline & 0 & & & I & \end{array}\right]$$

- if $\tilde{Q}_k = 0$, the factorization is complete ($r = k, P = P_k$)
- algorithm starts with $P_0 = I$ and $\tilde{Q}_0 = A$
- before step k, we reorder columns of $\tilde{Q}_{k-1}$ to place its largest column first
- this requires reordering columns $k, \ldots, n$ of R, and modifying P_{k-1}

Example

$$A = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & -1 \end{bmatrix}$$

Step 1

- a_2 and a_4 have the largest norms; we move a_2 to the first position
- find first column of Q, first row of R

$$\begin{aligned} \begin{bmatrix} a_2 & a_1 & a_3 & a_4 \end{bmatrix} &= \left[\begin{array}{c|ccc} 1/2 & 1/2 & 0 & 1 \\ 1/2 & -1/2 & 1 & -1 \\ 1/2 & 1/2 & 0 & 1 \\ 1/2 & -1/2 & -1 & -1 \end{array}\right] \left[\begin{array}{c|ccc} 2 & 1 & 0 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \\ &= \left[q_1 \mid \tilde{Q}_1\right] \left[\begin{array}{c|c} R_{11} & R_{1,2:4} \\ \hline 0 & I \end{array}\right] \end{aligned}$$

Example

Step 2

- move column 3 of $\tilde{Q}_1$ to first position in $\tilde{Q}_1$

$$\begin{bmatrix} a_2 & a_4 & a_1 & a_3 \end{bmatrix} = \left[\begin{array}{c|ccc} 1/2 & 1 & 1/2 & 0 \\ 1/2 & -1 & -1/2 & 1 \\ 1/2 & 1 & 1/2 & 0 \\ 1/2 & -1 & -1/2 & -1 \end{array}\right] \left[\begin{array}{c|ccc} 2 & 0 & 1 & 0 \\ \hline 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

- find second column of Q, second row of R

$$\begin{bmatrix} a_2 & a_4 & a_1 & a_3 \end{bmatrix} = \left[\begin{array}{cc|cc} 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 0 & 1 \\ 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 0 & -1 \end{array}\right] \left[\begin{array}{cc|cc} 2 & 0 & 1 & 0 \\ 0 & 2 & 1 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{cc|c} q_1 & q_2 & \tilde{Q}_2 \end{array}\right] \left[\begin{array}{cc|c} R_{11} & R_{12} & R_{1,3:4} \\ 0 & R_{22} & R_{2,3:4} \\ \hline 0 & 0 & I \end{array}\right]$$

Example

Step 3

- move column 2 of $\tilde{Q}_2$ to first position in $\tilde{Q}_2$

$$\begin{bmatrix} a_2 & a_4 & a_3 & a_1 \end{bmatrix} = \left[\begin{array}{cc|cc} 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & -1 & 0 \end{array}\right] \left[\begin{array}{cc|cc} 2 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

- find third column of Q, third row of R

$$\begin{bmatrix} a_2 & a_4 & a_3 & a_1 \end{bmatrix} = \left[\begin{array}{ccc|c} 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 1/\sqrt{2} & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & -1/\sqrt{2} & 0 \end{array}\right] \left[\begin{array}{ccc|c} 2 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & \sqrt{2} & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} q_1 & q_2 & q_3 & \tilde{Q}_3 \end{array}\right] \left[\begin{array}{ccc|c} R_{11} & R_{12} & R_{13} & R_{14} \\ 0 & R_{22} & R_{23} & R_{24} \\ 0 & 0 & R_{33} & R_{34} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]$$

Example

Result: since $\tilde{Q}_3$ is zero, the algorithm terminates with the factorization

$$\begin{bmatrix} a_2 & a_4 & a_3 & a_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & -1 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 1/\sqrt{2} \\ 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & \sqrt{2} & 0 \end{bmatrix}$$

Full pivoted QR factorization

any $A \in \mathbb{R}^{m\times n}$ admits the full pivoted QR factorization:

$$A = \begin{bmatrix} Q & Q_0 \end{bmatrix} \begin{bmatrix} R_1 & R_2 \\ 0 & 0 \end{bmatrix} P^T$$

- Q, $R = [R_1 \mid R_2]$ ($R_2 \in \mathbb{R}^{r\times(n-r)}$), and permutation P, are same as page 6.21
- $[Q \;\; Q_0]$ is an $m \times m$ orthogonal matrix
- columns of Q_0 are an orthonormal basis for $\mathrm{null}(A^T)$:

  $$\mathrm{range}(Q_0) = \mathrm{range}(A)^\perp = \mathrm{null}(A^T)$$

  this follows from

  $$A^T = P[R^T \;\; 0] \begin{bmatrix} Q^T \\ Q_0^T \end{bmatrix}$$

  and

  $$A^T z = PR^T Q^T z = 0 \quad \Longleftrightarrow \quad Q^T z = 0 \quad \Longleftrightarrow \quad z \in \mathrm{range}(Q_0)$$

- called *rank-revealing QR factorization* (RRQR)

Outline

- QR factorization
- QR via Gram-Schmidt
- modified Gram-Schmidt
- pivoted QR factorization
- **Householder algorithm**

Householder algorithm

- the most widely used algorithm for QR factorization (`qr` in MATLAB and Julia)
- less sensitive to rounding error than Gram-Schmidt algorithm
- computes a "full" QR factorization (QR decomposition)

$$A = \begin{bmatrix} Q & Q_0 \end{bmatrix} \begin{bmatrix} R \\ 0 \end{bmatrix}, \quad [Q \;\; Q_0] \text{ orthogonal}$$

  for matrix with linearly independent columns

  – can be modified to compute full pivoted QR factorization

- the full Q-factor is constructed as a product of orthogonal matrices

$$[Q \;\; Q_0] = H_1 H_2 \cdots H_n$$

  each H_i is an $m \times m$ symmetric and orthogonal

Reflector

Reflector: an *elementary reflector* is a matrix of the form

$$H = I - 2vv^T \quad \text{with } v \text{ a unit-norm vector } \|v\| = 1$$

Properties

- a reflector matrix is symmetric, and orthogonal

$$H^T H = (I - 2vv^T)(I - 2vv^T) = I - 4vv^T + 4vv^T vv^T = I$$

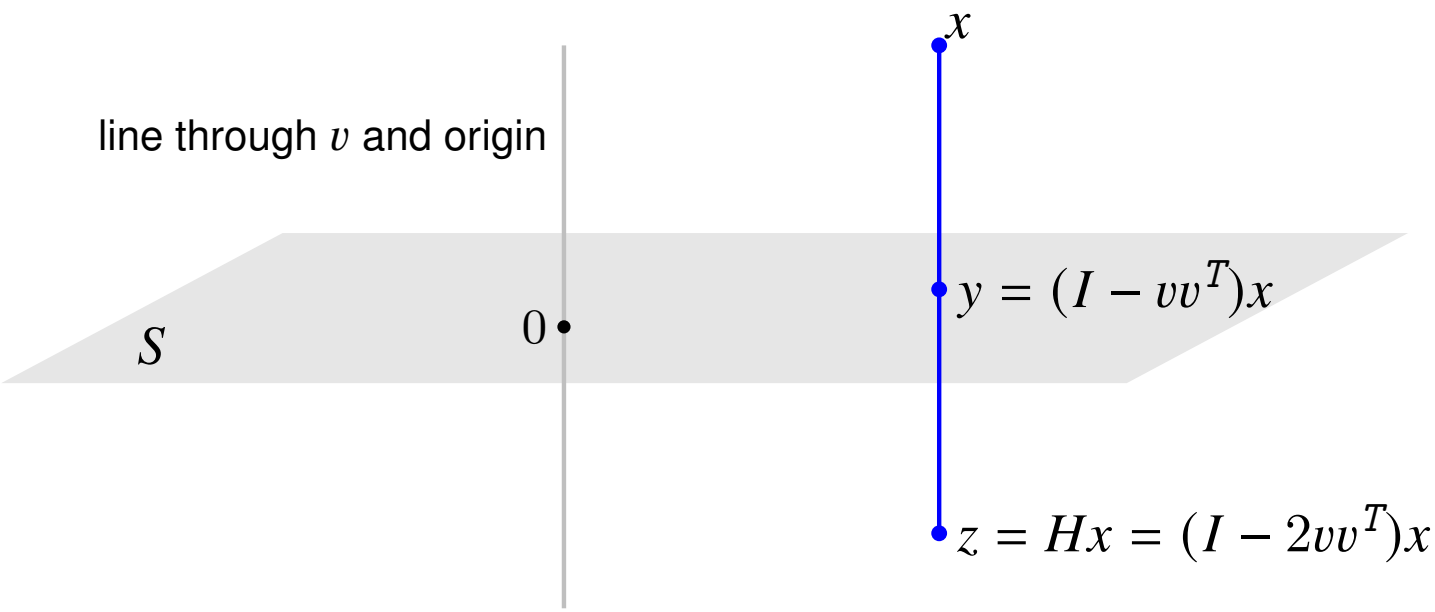
- reflection of v: $Hv = -v$

- matrix-vector product Hx can be computed efficiently as

$$Hx = x - 2(v^T x)v$$

  complexity is $4p$ flops if v and x have length p

Geometrical interpretation of reflector



- $S = \{u \mid v^T u = 0\}$ is the (hyper-)plane of vectors orthogonal to v
- if $\|v\| = 1$, the projection of x on S is given by

$$y = (I - vv^T)x$$

- reflection of x through the hyperplane is given by product with reflector:

$$z = y + (y - x) = (I - 2vv^T)x$$

Reflection to multiple of first unit vector

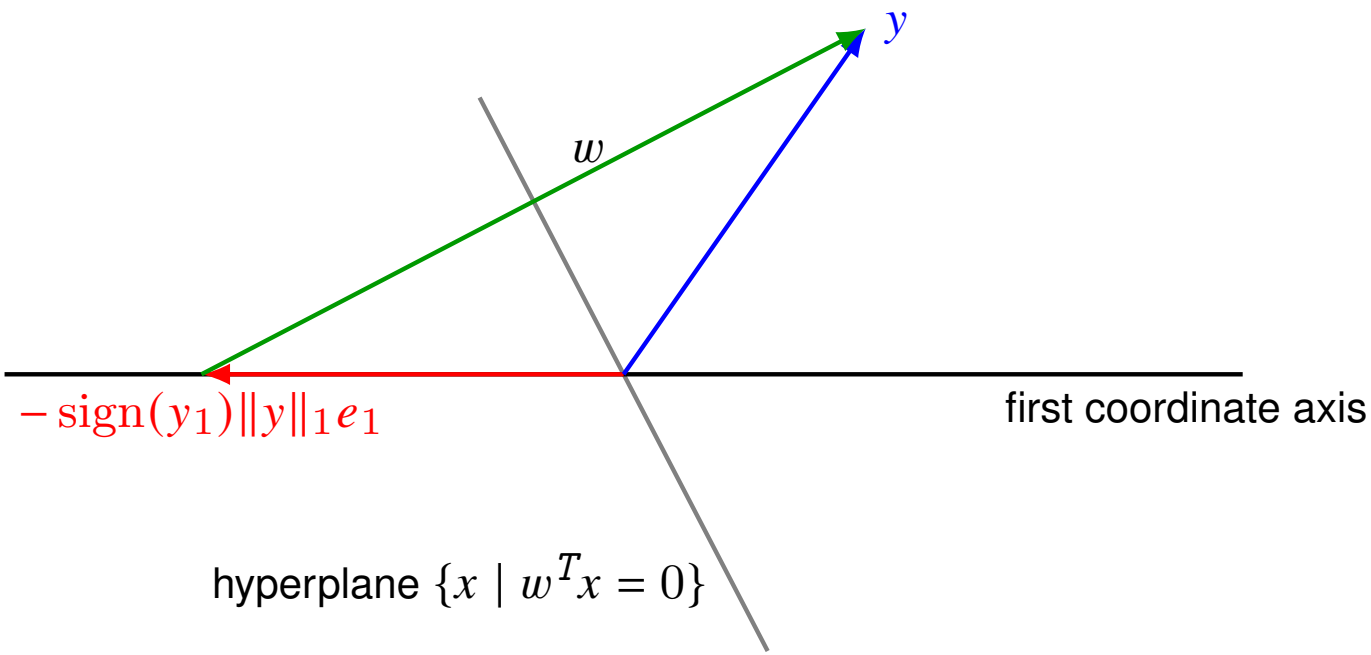
given nonzero p-vector $y = (y_1, y_2, \ldots, y_p)$, define

$$w = \begin{bmatrix} y_1 + \mathrm{sign}(y_1)\|y\| \\ y_2 \\ \vdots \\ y_p \end{bmatrix}, \quad v = \frac{1}{\|w\|} w$$

- we define $\mathrm{sign}(0) = 1$
- vector w satisfies
$$\|w\|^2 = 2(w^T y) = 2\|y\|(\|y\| + |y_1|)$$
- reflector $H = I - 2vv^T$ maps y to multiple of $e_1 = (1, 0, \ldots, 0)$:

$$Hy = y - \frac{2(w^T y)}{\|w\|^2} w = y - w = -\mathrm{sign}(y_1)\|y\| e_1$$

Geometry



the reflection through the hyperplane $\{x \mid w^T x = 0\}$ with normal vector

$$w = y + \operatorname{sign}(y_1)\|y\|e_1$$

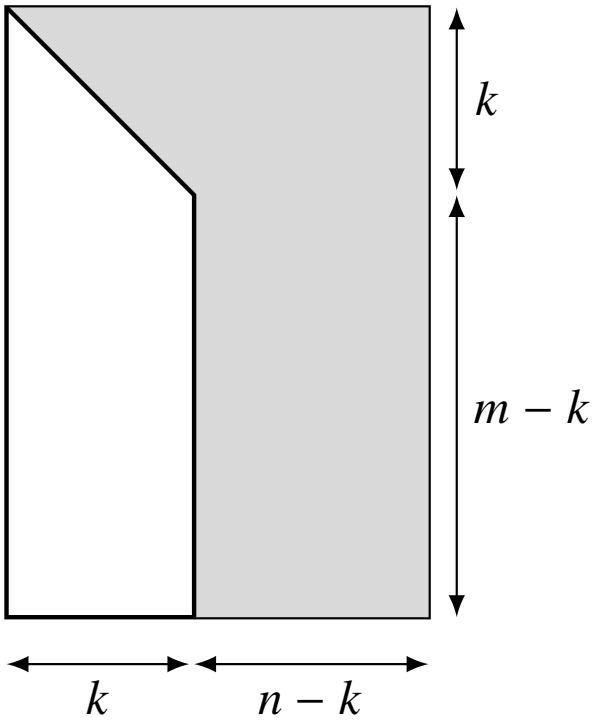
maps y to the vector $-\operatorname{sign}(y_1)\|y\|e_1$

Householder triangularization

- computes reflectors $H_1, \ldots, H_n$ that reduce A to triangular form:

$$H_n H_{n-1} \cdots H_1 A = \begin{bmatrix} R \\ 0 \end{bmatrix}$$

- after step k, the matrix $H_k H_{k-1} \cdots H_1 A$ has the following structure:



(elements in positions i, j for $i > j$ and $j \leq k$ are zero)

Householder algorithm

---

given: $m \times n$ matrix A with linearly independent columns $a_1, \ldots, a_n$

for $k = 1, 2, \ldots, n$

1. define $y = A_{k:m,k}$ and compute $(m - k + 1)$-vector v_k:

$$w = y + \mathrm{sign}(y_1)\|y\|e_1, \quad v_k = \frac{1}{\|w\|}w$$

2. multiply $A_{k:m,k:n}$ with reflector $I - 2v_k v_k^T$:

$$A_{k:m,k:n} := A_{k:m,k:n} - 2v_k(v_k^T A_{k:m,k:n})$$

- algorithm overwrites A with $\begin{bmatrix} R \\ 0 \end{bmatrix}$
- **complexity:** $2mn^2 - \frac{2}{3}n^3$ flops (we take $2mn^2$ for the complexity)

Remarks

- step 2 is equivalent to multiplying A with $m \times m$ reflector

$$H_k = \begin{bmatrix} I & 0 \\ 0 & I - 2v_k v_k^T \end{bmatrix} = I - 2 \begin{bmatrix} 0 \\ v_k \end{bmatrix} \begin{bmatrix} 0 \\ v_k \end{bmatrix}^T$$

- algorithm returns the vectors $v_1, \ldots, v_n$, with v_k of length $m - k + 1$

Q-factor

$$\begin{bmatrix} Q & Q_0 \end{bmatrix} = H_1 H_2 \cdots H_n$$

- usually there is no need to compute the matrix $[Q \;\; Q_0]$ explicitly
- the vectors $v_1, \ldots, v_n$ are an economical representation of $[Q \;\; Q_0]$
- products with $[Q \;\; Q_0]$ or its transpose can be computed as

$$\begin{bmatrix} Q & Q_0 \end{bmatrix} x = H_1 H_2 \cdots H_n x$$
$$\begin{bmatrix} Q & Q_0 \end{bmatrix}^T y = H_n H_{n-1} \cdots H_1 y$$

Example

$$A = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix} = H_1 H_2 H_3 \begin{bmatrix} R \\ 0 \end{bmatrix}$$

we compute reflectors H_1, H_2, H_3 that triangularize A:

$$H_3 H_2 H_1 A = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ 0 & 0 & R_{33} \\ 0 & 0 & 0 \end{bmatrix}$$

First column of R

- compute reflector that maps first column of A to multiple of e_1:

$$y = \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \quad w = y - \|y\| e_1 = \begin{bmatrix} -3 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \quad v_1 = \frac{1}{\|w\|} w = \frac{1}{2\sqrt{3}} \begin{bmatrix} -3 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

- overwrite A with product of $I - 2v_1 v_1^T$ and A

$$A := (I - 2v_1 v_1^T) A = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4/3 & 8/3 \\ 0 & 2/3 & 16/3 \\ 0 & 4/3 & 20/3 \end{bmatrix}$$

Second column of R

- compute reflector that maps $A_{2:4,2}$ to multiple of e_1:

$$y = \begin{bmatrix} 4/3 \\ 2/3 \\ 4/3 \end{bmatrix}, \quad w = y + \|y\| e_1 = \begin{bmatrix} 10/3 \\ 2/3 \\ 4/3 \end{bmatrix}, \quad v_2 = \frac{1}{\|w\|} w = \frac{1}{\sqrt{30}} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix}$$

- overwrite $A_{2:4,2:3}$ with product of $I - 2v_2 v_2^T$ and $A_{2:4,2:3}$:

$$A := \begin{bmatrix} 1 & 0 \\ 0 & I - 2v_2 v_2^T \end{bmatrix} A = \begin{bmatrix} 2 & 4 & 2 \\ 0 & -2 & -8 \\ 0 & 0 & 16/5 \\ 0 & 0 & 12/5 \end{bmatrix}$$

Third column of R

- compute reflector that maps $A_{3:4,3}$ to multiple of e_1:

$$y = \begin{bmatrix} 16/5 \\ 12/5 \end{bmatrix}, \quad w = y + \|y\| e_1 = \begin{bmatrix} 36/5 \\ 12/5 \end{bmatrix}, \quad v_3 = \frac{1}{\|w\|} w = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

- overwrite $A_{3:4,3}$ with product of $I - 2v_3 v_3^T$ and $A_{3:4,3}$:

$$A := \begin{bmatrix} I & 0 \\ 0 & I - 2v_3 v_3^T \end{bmatrix} A = \begin{bmatrix} 2 & 4 & 2 \\ 0 & -2 & -8 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

Final result

$$
\begin{aligned}
H_3H_2H_1A &= \begin{bmatrix} I & 0 \\ 0 & I - 2v_3v_3^T \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & I - 2v_2v_2^T \end{bmatrix} (I - 2v_1v_1^T)A \\
&= \begin{bmatrix} I & 0 \\ 0 & I - 2v_3v_3^T \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & I - 2v_2v_2^T \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 4/3 & 8/3 \\ 0 & 2/3 & 16/3 \\ 0 & 4/3 & 20/3 \end{bmatrix} \\
&= \begin{bmatrix} I & 0 \\ 0 & I - 2v_3v_3^T \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & -2 & -8 \\ 0 & 0 & 16/5 \\ 0 & 0 & 12/5 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 4 & 2 \\ 0 & -2 & -8 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix}
\end{aligned}
$$

References and further readings

- S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares.* Cambridge University Press, 2018.
- L. Vandenberghe, *EE133A Lecture Notes*, University of California, Los Angeles.
- L. Vandenberghe. *EE133B Lecture Notes*, University of California, Los Angeles.
- M. T. Heath. *Scientific Computing: An Introductory Survey* (revised second edition). Society for Industrial and Applied Mathematics, 2018.
- U. M. Ascher. *A First Course on Numerical Methods*. Society for Industrial and Applied Mathematics, 2011.